Agus Fuad Mudhofar

Institut Teknologi Sepuluh Nopember, Surabaya

+62-8882111890
agusfuad090@gmail.com
GitHub Profile
LinkedIn Profile

Education

•**Bachelor of Computer Engineering** *2022–Present*
Institut Teknologi Sepuluh Nopember, Surabaya GPA: 3.64

Experience

•**Research Intern – Mobile Developer & Computer Vision Engineer** *Feb – July 2026*
Bandung Badan Riset dan Inovasi Nasional (BRIN)

a. Full-Stack & Application Development
   - -Built a cross-platform Flutter app (Mobile, Tablet, Desktop) to calculate NPK nutrient requirements for vegetable crops.
   - -Engineered a RESTful API backend with Laravel 12 and PostgreSQL, implementing Laravel Sanctum for security.

b. AI & Computer Vision Systems
   - -Developed a situational awareness system utilizing camera input, deep learning models, and LLM integration to interpret activities (walking, driving) and recognize human actions.

•**IoT Developer – Smart Wheelchair Development** *Aug 2024 – Present*
Surabaya Multimedia and IoT Laboratory (MIOT)

– Maintained and enhanced a smart wheelchair system utilizing deep learning-based computer vision for hands-free navigation.
– Integrated real-time eye-gaze and facial gesture recognition using Python and OpenCV, improving system responsiveness.
– Developed custom microcontroller firmware (C/C++) for motor actuation and sensor feedback, ensuring safe navigation.

•**Research Assistant – Multimedia and IoT Lab (B300)** *April – June 2025*
Surabaya Multimedia and IoT Laboratory (MIOT)

– Conducted research on deploying deep learning models on edge devices (NVIDIA Jetson Nano) for computer vision tasks.
– Developed object detection pipelines (Python/C++) using MobileNet-SSD, YOLOv5n, and YOLOv11s architectures.
– Optimized model inference using ONNX and NVIDIA TensorRT, and integrated them into NVIDIA DeepStream SDK.

•**Lab Teaching Assistant – Telemathics Workshop** *Mar 2025 – May 2025*
Surabaya Robotics and Digital Systems Laboratory (B401)

– Supervised students in hardware prototyping: manual soldering, hot air rework, reflow soldering, and 3D printing.
– Provided technical assistance in C/C++ programming and microcontroller-based hardware-software troubleshooting.

Projects

•**Smart Wheelchair with Eye Movement Control**
A smart wheelchair system controlled by eye movement, integrating computer vision and embedded hardware.

– Designed and prototyped a smart wheelchair enabling users to control movement using eye gaze and facial gestures.
– Integrated computer vision (Python/OpenCV) with microcontroller (Arduino/ESP32) actuation and sensor feedback.
– Collaborated with a team to showcase the functional prototype at the GEMASTIK XVII competition.
– Technology Used: Python, OpenCV, Arduino/ESP32, C/C++, Embedded Hardware.

- **Dot Matrix Ping Pong Game (Embedded System)**

  A microcontroller-based two-player Ping Pong game using an 8x32 LED dot matrix and custom C++ firmware.

  – Implemented game mechanics, scorekeeping, and animations on an 8x32 LED matrix using MD_Parola and MD_MAX72xx.
  – Designed circuit schematics featuring analog paddle control (potentiometers) and digital smash buttons via SPI.
  – Technology Used: C++ (Arduino/ESP32), SPI, MD_Parola, MD_MAX72xx, Circuit Design.

- **Sappy: Multi-Role Livestock Management App**

  A livestock management app for multiple roles (Farmer, Admin, Doctor) with NFC tracking support.

  – Implemented state management via Provider and enabled NFC functionality for secure livestock identification and tracking.
  – Technology Used: Flutter, Dart, NFC, Provider, fl_chart, carousel_slider.

Positions of Responsibility

- **Team Leader – GEMASTIK XVII (IoT Division)** *Jan – Nov 2024*

  Semarang Semarang State University (UNNES)

  – Led a team representing ITS in the Smart Devices, Embedded Systems, and IoT Division at GEMASTIK XVII.
  – Coordinated software and hardware components, managed timelines, and secured the National Finalist (Juara Harapan) award.

- **PIC IoT Division – TIM Kawal GEMASTIK** *Jan – Nov 2025*

  Surabaya Institut Teknologi Sepuluh Nopember (ITS)

  – Coordinated communication between faculty supervisors, professors, mentors, and participating student teams in the IoT division.
  – Evaluated proposals and technical readiness, and organized mentoring sessions with IoT domain experts.

Certifications

- **English Proficiency Certification – CEFR Level B2** *2024*

  DynEd International

  – Achieved English proficiency equivalent to the Common European Framework of Reference for Languages (CEFR) Level B2.

Technical Skills and Interests

Languages: C/C++, Python, Dart, Javascript, HTML+CSS
Hardware & IoT: Arduino, ESP32, NVIDIA Jetson Nano, Raspberry Pi, Microcontrollers, SPI, I2C, UART, Soldering, 3D Printing, PCB/Circuit Design
Frameworks & Libraries: OpenCV, Flutter, React.js, Laravel, FastAPI, TensorRT, ONNX, DeepStream SDK
Databases & Cloud: PostgreSQL, MySQL, MongoDB, Firebase
Relevant Coursework: Embedded Systems, Microprocessors, Digital Systems, Computer Networks, Data Structures & Algorithms
Languages Info: Indonesian (Native), English (B2 CEFR - DynEd Certified)
Soft Skills: Problem Solving, Team Collaboration, Technical Communication, Adaptability, Self-learning